

A Large Vision-Language Model Based System for Detecting Image-Text Misinformation

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Abstract: **A real news photograph can be reused with a caption from a different event. Nothing in the image is edited, so image forensics will not catch it, and the caption on its own may look perfectly reasonable. Detecting this kind of out-of-context misinformation means judging whether the picture and the text describe the same event. In this study I built a detector on LLaVA-1.5-7B, compressed to 4 bits and fine-tuned with LoRA on a single free GPU. It follows a two-stage design (news-domain tuning, then task tuning) and was tested with and without web-evidence retrieval and generated explanations. I read each verdict as a token probability and set the decision threshold on validation data. Training used COSMOS, and MMFakeBench served as an unseen test set. The final system, the base model with news-domain tuning, scores 83.2% accuracy, 0.832 macro F1 and 0.910 AUROC on held-out COSMOS data, and it improves significantly on the untrained model on both benchmarks. Reading the same untrained model's answer as free text gives only 50.0%, against 78.5% with calibrated scoring. Leakage checks, significance tests, baselines and an image ablation support the main result, but they also show that detection depends mostly on the captions. Task tuning on 1,189 examples and an integrated retrieval pipeline did not improve the system. A small open model can therefore come within two points of the original COSMOS method, provided its answers are read as calibrated probabilities.**

Index Terms: **misinformation detection, out-of-context, vision-language models, LLaVA, LoRA, calibration, retrieval augmentation**

I. Introduction

A. Background Information

Picture a photograph of a flooded street from 2015, shared in 2024 with a caption claiming it shows last night's storm. The photo is real and so is the flood. Only the pairing is false. Over the past decade a large body of research has looked at how false content like this spreads, and how quickly it reaches people before any fact-checker can respond (Chen and Shu, 2024; Patra et al., 2022). Posts that combine an image with text are particularly convincing, because people tend to trust what they can see.

This out-of-context case is hard to catch automatically. There are no editing artefacts for image forensics to find, and a text-only fact-checker sees a fluent, plausible caption. The only way to detect it is to ask whether the image and the caption belong together (Aneja et al., 2023; Luo et al., 2021).

Large vision-language models (LVLMs) read images and text at the same time, which makes them a natural fit. Recent systems tune them on news data (Qi et al., 2024), give them retrieved web evidence (Xuan et al., 2024; Tahmasebi et al., 2024), or break the decision into smaller questions (Liu et al., 2025). Most of these systems use 13B-parameter models or GPT-4V, and several are trained on tens of thousands of instructions written by GPT-4.

B. Research Problem and Question

It remains unclear whether a small open model, trained on the kind of hardware a student or a small newsroom has, can do this reliably. It is also unclear which of the components proposed in the literature actually help. Two further issues motivated this study. Published LVLM results are usually obtained by parsing the model's generated answer, and nobody seems to have checked how much that choice affects the measured score. And COSMOS, a standard benchmark for this task, is known to be largely solvable from the text alone (Papadopoulos et al., 2023), yet LVLM detectors are rarely tested for whether they use the image at all.

The research question is:

Can a multimodal misinformation detector built on a domain-adapted LVLM with retrieval augmentation achieve accurate, generalisable detection with useful explanations, and which components actually contribute?

The purpose of this study was to build such a detector around LLaVA-1.5-7B on a free GPU, measure what news-domain tuning, task tuning, retrieval and explanation generation each add, and then test whether the result holds up against leakage, chance, simple baselines and a benchmark the model never saw.

C. Significance of the Research

A detector is only useful to fact-checkers if it is cheap enough to run and its reported accuracy can be trusted. The final system here reaches 83.2% accuracy and 0.910 AUROC on COSMOS, within two points of the original COSMOS method. Just as important is a measurement finding. When I parsed the untrained model's written answers it scored 50.0%, no better than a coin. When I read its probability for the verdict and set the threshold on validation data, the same model scored 78.5%. That gap is larger than anything training achieved, and it suggests that free-text evaluation may understate what LVLMs can do on this task.

The results also show where the approach falls short. Detection relies mainly on the captions. Task tuning on a small labelled set, and retrieval built into the verdict prompt, both made the system worse.

Section II reviews related work and Section III sets out the methodology. Section IV reports the results and the validation experiments, Section V discusses them against prior work, and Section VI concludes.

II. Literature Review

A. Overview of Relevant Literature

1) Pre-LVLM approaches: Early fake news detectors worked mostly with text: writing style, word choice and how a post spread through a network (Patra et al., 2022). COSMOS (Aneja et al., 2023) moved the focus to images. It is not a language model. It uses a frozen, pre-trained Mask R-CNN (He et al., 2017) with a ResNet-50 backbone (He et al., 2016) to find objects in each photo, and the frozen Universal Sentence Encoder (Cer et al., 2018) to embed each caption. A small network on top is trained, without any out-of-context labels, on about 160,000 news images, so that a photo's real caption matches one of its objects better than a random caption does. At test time a fixed rule decides: if both captions point to overlapping objects (IoU above 0.5) but SBERT (Reimers and Gurevych, 2019) finds the captions dissimilar (similarity below 0.5), the image is out-of-context. This reaches 85% accuracy on COSMOS's 1,700 labelled test images. The SBERT caption comparison on its own already gets 77%. KDCN (Sun et al., 2023) checks two kinds of consistency at once, between image and text and between the post and a knowledge graph. NewsCLIPpings (Luo et al., 2021) generated out-of-context pairs automatically at scale. Its CLIP (Radford et al., 2021) classifier reached 60.2%, and human annotators managed only 65.6%, which says a lot about how hard the task is.

2) LVLM-based detection: SNIFFER (Qi et al., 2024) is the closest design to mine. It tunes InstructBLIP with a Vicuna-13B language model in two stages: first on news image descriptions, then on 35,536 out-of-context instructions generated by GPT-4. It combines an internal check of the image and caption with an external check against retrieved evidence, and reports 88.4% on NewsCLIPpings. Interestingly, the evidence adds only 0.2 points.

LEMMA (Xuan et al., 2024) uses GPT-4V without fine-tuning. The model makes a first judgement, writes search queries, gathers and filters evidence, and then revises its verdict. This lifts accuracy to 82.4% on Twitter and 82.8% on Fakeddit (Nakamura et al., 2020), roughly 7 points above GPT-4V on its own. Tahmasebi et al. (2024) use LVLMs both to re-rank evidence and to verify claims. With LLaVA-1.6-7B, retrieved evidence produced an F1 of 0.451 on MOCHEG, below the 0.491 baseline; gold evidence raised it to 0.534. FKA-Owl (Liu et al., 2024b) takes another route and gives the LVLM forgery-specific knowledge.

3) Benchmarks, explanations and evaluation: MMFakeBench (Liu et al., 2025) mixes three kinds of forgery: false text, edited images and mismatched pairs. Its results for open models are low. Under standard prompting, LLaVA-1.6-7B scores 19.0 four-class macro F1 on the test set. The authors' MMD-Agent splits detection into separate questions about the text, the image and their consistency, which brings LLaVA-1.6-34B up to 47.7 and GPT-4V to 61.5.

On the evaluation side, Mocheg (Yao et al., 2023) scores explanations as well as verdicts. Thibault et al. (2024) audited misinformation datasets, found many with shortcuts a model can exploit, and recommend simple baselines and reading the outputs by hand. VERITE (Papadopoulos et al., 2023) found that on COSMOS a text-only model (72.6%) beats multimodal ones. Recent surveys (Ai et al., 2026; Chen and Shu, 2024) still list fine-grained alignment between image and text as an open weakness.

B. Key Theories and Concepts

LLaVA (Liu et al., 2023, 2024a) connects a CLIP image encoder to a language model and is trained to follow instructions about pictures. Tuning such a model further on domain or task instructions, as SNIFFER does, is called visual instruction tuning.

Fine-tuning all 7 billion parameters is out of reach on a free GPU. LoRA (Hu et al., 2022) avoids this by freezing the model and training small low-rank update matrices, one of which starts at zero. QLoRA (Dettmers et al., 2023) applies the same idea to a model stored in 4-bit precision, which is what makes a 7B model trainable in 16 GB of memory.

The verdict in this study is read from token probabilities. A language model's probability for its next token says something about whether its answer is right (Kadavath et al., 2022). Those probabilities are also biased towards certain answers

(Zhao et al., 2021), so they have to be adjusted before use. How well probabilities match observed frequencies is called calibration. It is usually measured with expected calibration error (ECE) and corrected with simple post-hoc methods such as Platt scaling (Guo et al., 2017; Platt, 1999).

Turning a probability into a yes or no needs a threshold. The threshold that gives the best F1 is usually not 0.5 and has to be chosen from the data (Lipton et al., 2014). AUROC avoids the choice altogether: it measures how well the scores separate the two classes across every possible threshold, and it is not affected by class balance (Fawcett, 2006).

Finally, a benchmark has unimodal bias when one modality alone is enough to solve it. A model can then score well without ever relating the image to the text (Papadopoulos et al., 2023; Thibault et al., 2024).

C. Gaps in the Literature

Gap 1: small open models. SNIFFER and LEMMA rely on 13B models or GPT-4V and on large generated instruction sets. Nobody has tested whether their designs carry over to a 7B open model with roughly a thousand labelled examples.

Gap 2: how answers are read. LVLM results, including those on MMFakeBench, come from parsing generated text. Given the known answer bias of language models (Zhao et al., 2021), this may understate them, but the effect has not been measured for misinformation detection.

Gap 3: what the detector actually uses. Few LVLM detectors are checked for unimodal shortcuts or for overlap between training and test data, even though both are known problems (Papadopoulos et al., 2023; Kapoor and Narayanan, 2023).

Gap 4: generalisation and explanations. Performance on mixed-source forgeries is still weak (Liu et al., 2025), and explanations are seldom evaluated alongside accuracy (Yao et al., 2023).

III. Methodology

A. Research Design

The study is experimental and quantitative. I built the detector one component at a time and evaluated every version on the same held-out data, so that each component's contribution could be measured directly. Fig. 1 shows the overall system.

1) Model and fine-tuning: The base model is LLaVA-1.5-7B (Liu et al., 2023, 2024a), loaded with 4-bit NF4 quantisation. I trained LoRA adapters (Hu et al., 2022) on top of the frozen model in the QLoRA style (Dettmers et al., 2023), with rank 16, alpha 32 and dropout 0.05. Training used a OneCycle learning-rate schedule and an effective batch size of 16 (4 per step, accumulated over 4 steps). Everything ran on Kaggle's free NVIDIA T4 GPUs. The quota is 30 GPU-hours a week and a session stops after 12 hours, so I saved checkpoints to persistent storage throughout training.

2) Stage 1: news-domain tuning: Following SNIFFER (Qi et al., 2024), Stage 1 teaches the model to describe COSMOS news images using their real captions. The aim was to get it naming the specific people and places in news photos instead of saying "a man in a suit". For each run I kept the checkpoint with the lowest validation loss.

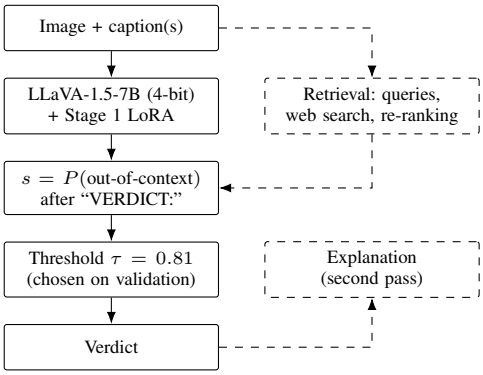


Fig. 1. System overview. Dashed components were evaluated but are not part of the final detector (retrieval) or do not affect the verdict (explanation).

3) Stage 2: task tuning: Stage 2 adds a second LoRA adapter, trained for three epochs on the labelled examples. The target is the verdict followed by a short reasoning sentence. After finding a name collision between the two adapters (Section IV-A1), I restricted Stage 2 to the attention layers of the language model and left the vision encoder out.

4) Retrieval: The retrieval step follows LEMMA (Xuan et al., 2024). The model writes a first assessment, generates search queries from the captions and that assessment, and runs a web search. It then re-ranks the results (Tahmasebi et al., 2024), keeps the top three snippets, and gives a final verdict from a prompt containing the captions, its first assessment and the evidence. I cached all the evidence so the final prompt could be re-scored with parts removed. I also tried SNIFFER's approach of keeping the internal and external checks separate (Qi et al., 2024). Here I combined the plain verdict score and the evidence-based score with a logistic regression fitted on validation data, much like Platt scaling (Platt, 1999).

5) Explanations: Explanations come from a second pass. The base model sees the image, the captions and the verdict the final system gave, and is asked to explain the visual and textual evidence for it. The explanation is written after the decision is made, so it cannot change the detection results.

B. Data Collection Methods

The data used for this study were collected from three public benchmarks. No data were collected from people.

I first planned to use NewsCLIPpings as the main dataset. Its images, however, are distributed as one 91 GB archive, and after several days of attempts, including two corrupted downloads, I could not get a complete copy. With my supervisor's approval I switched to COSMOS (Aneja et al., 2023), which addresses the same task and is distributed in smaller files. COSMOS's training and validation splits contain unlabelled image-caption pairs scraped from news and fact-checking sites. Only the test split is labelled: each image there has two captions, and the label says whether using both with that image is out-of-context. While preparing the data I found

TABLE I
COSMOS SPLITS USED

Split	Size
Unlabelled training (Stage 1)	161,752 images
Unlabelled validation (Stage 1)	41,006 images
Labelled training (Stage 2)	1,189 examples
Labelled validation (threshold, model selection)	254 examples
Labelled test (held out)	256 examples

TABLE II
METHODOLOGICAL DECISIONS AND SUPPORTING LITERATURE

Decision	Supported by
LLaVA-1.5-7B base model	(Liu et al., 2024a, 2025)
4-bit quantisation with LoRA	(Hu et al., 2022; Dettmers et al., 2023)
Two-stage tuning (domain, then task)	(Qi et al., 2024)
COSMOS for training; MMFakeBench for evaluation only	(Aneja et al., 2023; Liu et al., 2025)
Token probability instead of free text	(Kadavath et al., 2022; Zhao et al., 2021; Thibault et al., 2024)
Threshold by macro F1 on validation, never on test	(Lipton et al., 2014; Kapoor and Narayanan, 2023)
AUROC as primary measure	(Fawcett, 2006)
Paired bootstrap; Bonferroni correction	(Efron and Tibshirani, 1993; Koehn, 2004; Dunn, 1961)
Train-test leakage check	(Kapoor and Narayanan, 2023)
Blank-image ablation; simple baselines	(Papadopoulos et al., 2023; Thibault et al., 2024)
SBERT and CLIP baselines	(Reimers and Gurevych, 2019; Aneja et al., 2023; Radford et al., 2021; Luo et al., 2021)
Retrieval design; LVLM re-ranking	(Xuan et al., 2024; Tahmasebi et al., 2024)
Score fusion of internal and external checks	(Qi et al., 2024; Platt, 1999)
Decomposed questioning	(Liu et al., 2025)
ECE and Platt scaling	(Guo et al., 2017; Platt, 1999)
Explanation criteria; Mann-Whitney test	(Yao et al., 2023; Qi et al., 2024; Mann and Whitney, 1947)

that about 26% of the 360,758 raw training records were exact duplicates of the same image and caption, usually a wire-service caption reused by several outlets. I removed these, keeping distinct captions of the same image.

MMFakeBench (Liu et al., 2025) was used only for evaluation. Its test set has 10,000 items, each with a single caption: 3,000 real, 3,000 with false text, 3,000 mismatched pairs (the type closest to the COSMOS task) and 1,000 with edited images.

For NewsCLIPpings (Luo et al., 2021) I could still run a CLIP baseline, because the authors released precomputed CLIP embeddings. I used the full merged_balanced test split of 7,264 pairs.

C. Sample Selection

The labelled COSMOS data were split 70/15/15 into training, validation and test sets, stratified by label (Table I). The test set holds 128 out-of-context and 128 consistent items, and I used it only for the final evaluation.

For Stage 1, my first run used 10,000 unlabelled pairs sampled at random. A count of named entities then showed that 44.6% of the distinct names in the pool appear only once, so the model had little chance to learn most individuals. For the second run I sampled 12,000 pairs whose caption names a person mentioned at least three times. I found people with spaCy and grouped different forms of the same name by surname. This kept 51.4% of the training images, and the second run's adapter is the Stage 1 used everywhere in this paper.

On MMFakeBench, every evaluation splits the items, stratified by forgery type, into a calibration half and an evaluation half. I ran one configuration on all 10,000 items. To compare configurations within the GPU budget, I used an identical stratified sample of 2,000 items (seed 42) for all of them. For the explanation study I drew 50 test items, 25 from each class.

D. Data Analysis Techniques

1) Calibrated verdict scoring: Rather than parsing a generated answer, I run the model up to the point just after "VERDICT:" and read the probabilities of the first tokens of "out-of-context" (p_o) and "consistent" (p_c):

$$s = \frac{p_o}{p_o + p_c} \in [0, 1]. \tag{1}$$

These probabilities are informative (Kadavath et al., 2022) but biased (Zhao et al., 2021), so the cut-off matters. I swept thresholds from 0.05 to 0.95 in steps of 0.01 on the validation split and kept the one with the highest macro F1 (Lipton et al., 2014). The test set was never used for this (Kapoor and Narayanan, 2023). For the final system the threshold is 0.81.

2) Metrics and statistical tests: AUROC is the main measure, because it does not depend on the threshold or on class balance (Fawcett, 2006). I also report accuracy and macro F1. To test whether two configurations really differ, I used a paired bootstrap (Efron and Tibshirani, 1993; Koehn, 2004) with 5,000 resamples and 95% confidence intervals. Because I ran 22 tests in total, I applied a Bonferroni correction (Dunn, 1961). Calibration is measured with ECE over 15 bins (Guo et al., 2017), and explanation ratings are compared with a Mann-Whitney test (Mann and Whitney, 1947).

3) Validation experiments: To check the main result I ran a leakage analysis between training and test data, an evaluation on MMFakeBench, a blank-image ablation, SBERT (Reimers and Gurevych, 2019) and CLIP (Radford et al., 2021) baselines, a calibration analysis, MMD-Agent-style decomposed questions, a retrieval ablation with score fusion, and blind ratings of the explanations. Table II lists each methodological decision with the work that supports it.

IV. RESULTS

A. Presentation of Findings

1) Problems found during development: Three problems came up while building the system, and each one changed how later results were produced.

The first appeared in the zero-shot baseline. I let the untrained model write its verdict and parsed the text, and it called 248 of the 256 test items out-of-context. Accuracy was

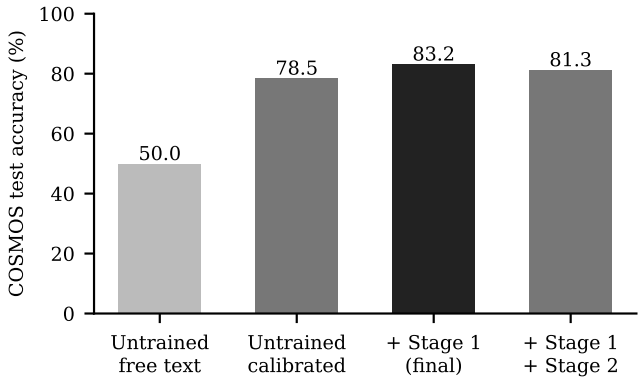


Fig. 2. COSMOS test accuracy by reading method and training stage ($n = 256$).

TABLE III
COSMOS TEST RESULTS WITH CALIBRATED SCORING ($n = 256$)

Configuration	τ	Val. F1	AUROC	Acc.	F1
Base LLaVA-1.5	0.85	0.791	0.875	0.785	0.785
+ Stage 1 (final)	0.81	**0.839**	**0.910**	**0.832**	**0.832**
+ Stage 1 + Stage 2	0.85	0.827	0.893	0.813	0.812

50.0% and macro F1 0.359. That looked like a model that could not do the task. As Section IV-B6 shows, the real cause was the reading method, and this is what led me to calibrated scoring.

The second was a silent bug in how the adapters were loaded. I trained Stage 2 for two more epochs and the new checkpoint gave exactly the same confusion matrix as the old one, down to four decimal places, even though the saved weights were different. Loading Stage 1 alone gave the same result again. At inference time the code merged Stage 1 into the base model before attaching Stage 2. Merging changes the internal parameter names, so the PEFT library could not match Stage 2's saved weights and quietly left every one at its zero starting value (Hu et al., 2022), without a warning. Attaching both adapters first and merging afterwards fixed it. A second, smaller collision on the vision encoder disappeared once Stage 2 was limited to the language model.

The third was simpler. The first retrieval run returned no evidence at all. The model was wrapping its search queries in quotation marks, which the search API treats as exact-phrase searches. Stripping the quotes fixed it.

2) Reading method and training stages: Fig. 2 and Table III give the COSMOS results. With calibrated scoring, the untrained model goes from 50.0% to 78.5% accuracy (AUROC 0.875). That 28.5-point jump comes without any training. Stage 1 then adds 4.7 points, reaching 83.2% and an AUROC of 0.910. Stage 2 makes things slightly worse.

3) Final system: I chose the Stage 1 configuration as the final system because it has the best validation macro F1 (0.839). The test results played no part in that choice. Its mistakes are evenly split. It flags 21 consistent pairs as out-of-context and misses 22 out-of-context ones (confusion matrix $[[107, 21], [22, 106]]$, rows are the truth). Precision and recall are 0.829 and 0.836 for consistent pairs and 0.835 and 0.828 for out-of-context pairs.

TABLE IV
PAIRED BOOTSTRAP TESTS (5,000 RESAMPLES)

Comparison	Data	ΔAUROC [95% CI]	p
Stage 1 vs base	COSMOS	+0.035 [+0.020, +0.053]	<0.001
Stage 1 vs base	MMFB	+0.028 [+0.015, +0.041]	<0.001
Stage 1+2 vs Stage 1	COSMOS	−0.017 [−0.032, −0.003]	0.013
Stage 1+2 vs Stage 1	MMFB	−0.006 [−0.018, +0.007]	0.387
Real vs blank, Stage 1	COSMOS	+0.014 [−0.011, +0.040]	0.293
Real vs blank, Stage 1+2	COSMOS	+0.058 [+0.027, +0.090]	<0.001

Stage 1 did not do what it was designed to do. I compared descriptions from the base and tuned models on 20 news images that name a person, once after each Stage 1 run. Both times the tuned model newly named only someone who was already very famous. It also made one new mistake, placing a photo from Australia in the United States.

B. Data Analysis and Interpretation

1) Statistical significance: Table IV gives the paired bootstrap results. Stage 1's gain over the base model is significant on both benchmarks ($p < 0.001$) and survives the Bonferroni correction, which needs p below about 0.002. Stage 2's drop on COSMOS ($p = 0.013$) does not survive it, so I treat it as suggestive.

2) Train-test leakage: COSMOS captions are often reused across news outlets, and overlap between training and test data is a well-known source of inflated results (Kapoor and Narayanan, 2023). I compared every test item with each training set in four ways: identical image files, near-identical images (a perceptual hash within a Hamming distance of 4), identical captions, and near-identical captions (word overlap of at least 0.8). The final system's training data shares nothing with the test set, and its validation data shares one near-duplicate caption.

Stage 2 was a different story. Its training data overlaps 126 of the 256 test items. The cause was my split. I divided the labelled records one by one, but the same image appears in several records with different captions, so one image could land in both the training and test sets. To see whether this changed any conclusion, I re-scored all three configurations on the 130 test items with no overlap. AUROC there is 0.861 for the base model, 0.907 for Stage 1 and 0.887 for Stage 1 plus Stage 2. The order is the same as before. Stage 2 still falls below Stage 1 even on items it could not have memorised.

3) Generalisation to MMFakeBench: Table V shows the results on the 2,000-item sample. None of this data was used for training, yet the pattern matches COSMOS. Stage 1 gives the highest AUROC (0.707) and macro F1, and Stage 2 adds nothing. So the Stage 1 gain is not a matter of memorising COSMOS. The full 10,000-item run with Stage 1 plus Stage 2 gave an AUROC of 0.698 and the same ordering by forgery type. Carrying the COSMOS threshold over unchanged cost

TABLE V
MMFAKEBENCH, AUROC ON 1,000 EVALUATION ITEMS (BY TYPE: VS. REAL)

Configuration	AUROC	Macro F1	Mismatch	Textual	Visual
Base LLaVA-1.5	0.679	0.625	0.568	0.803	0.638
+ Stage 1 (final)	**0.707**	**0.634**	0.590	0.823	0.706
+ Stage 1 + Stage 2	0.701	0.633	0.589	0.828	0.656
CLIP zero-shot	0.649		0.642	0.628	0.730

TABLE VI
CALIBRATION ON COSMOS TEST (ECE WITH 15 BINS)

Configuration	AUROC	ECE raw	Mean s	ECE Platt
Base	0.874	0.341	0.839	0.083
+ Stage 1 (final)	0.910	0.338	0.795	0.070
+ Stage 1 + 2	0.893	0.348	0.844	0.064

TABLE VII
RETRIEVAL ABLATION ON COSMOS ($n = 256$)

Final prompt contains	AUROC	Acc.
Standard verdict prompt (final system)	**0.910**	**0.832**
Retrieval template only	0.873	0.797
Template + evidence	0.862	0.746
Template + evidence + initial assessment	0.747	0.680
Final system fused with evidence score	0.923	0.840

only 0.02 to 0.04 macro F1 compared with re-calibrating on MMFakeBench.

The breakdown by forgery type was not what I expected. The model does best on false text (0.823), a type it was never trained on. It does worst on mismatched pairs (0.590), which is the type closest to its own task. On that type plain CLIP beats every LLaVA configuration I tried.

4) Contribution of the image: To see how much the system relies on the picture, I replaced every image with a plain grey one and recalibrated, a standard check for shortcuts (Papadopoulos et al., 2023; Thibault et al., 2024). The final system's COSMOS AUROC dropped from 0.910 to 0.896. That difference is not significant (Table IV). Working from the captions alone, it still beats the untrained model with the real images (0.875). Stage 1 plus Stage 2 leans on the image more (-0.058, significant), but it is the weaker system overall. On MMFakeBench the grey images had no effect on the edited-image category either, 0.656 with images against 0.657 without. Since the model spots these fakes just as well without seeing them, it must be picking up something in their captions.

5) Reference baselines: A text-only baseline that measures how different the two captions are (SBERT; Reimers and Gurevych, 2019) reaches 0.675 AUROC on COSMOS. The best CLIP (Radford et al., 2021) variant, which compares the image with each caption, reaches 0.633. The final system beats the combined CLIP baseline by 0.302 AUROC (95% CI [+0.227, +0.378], $p < 0.001$), so its performance cannot be explained by surface similarity. On NewsCLIPpings, zero-shot CLIP with a threshold set on validation data gets 65.3% accuracy (AUROC 0.720).

6) Calibration: The probabilities sort items well but are badly calibrated (Table VI). Exactly half the test items are out-of-context, yet the average predicted probability is between 0.80 and 0.84. For the final system every score falls between 0.47 and 0.93, a shift towards one answer of the kind Zhao et al. (2021) describe. A threshold of 0.5 therefore marks almost everything as out-of-context. That explains both the 50% free-text result and why the chosen threshold sits at 0.81. Platt scaling (Platt, 1999) fitted on the validation split brings ECE down from about 0.34 to between 0.06 and 0.08, in line with Guo et al. (2017). All three configurations have almost the same raw ECE (0.338 to 0.348), even though their AUROC differs.

7) Decomposed questioning: Following MMD-Agent (Liu et al., 2025), I also asked the model separate questions about each MMFakeBench item. Is the text false? Has the image been edited? Do they match? I added one four-way question as well, and read every answer as a probability. Simply taking the highest of the three check scores did worse than the single verdict (-0.027 AUROC with Stage 1, $p = 0.039$), because the checks produce scores on different scales. Combining them with a logistic regression fitted on the calibration half worked much better. With Stage 1 it reached 0.739 AUROC ($+0.032$, $p = 0.004$) and 58.5 four-class macro F1; the base model reached 0.730 and 55.3.

The visual check was the surprise. It separated edited images from real ones almost perfectly, with an AUROC of 0.974. So the model can use the image when the question is about the image. Stage 1 improved all three checks, although none of them was part of its training. Mismatch detection stayed around 0.61.

8) Retrieval and fusion: Table VII removes the parts of the retrieval prompt one at a time. With everything included, AUROC falls from 0.910 to 0.747 ($p < 0.001$). Taking out the model's first assessment recovers 0.115 of that. The first assessment is generated as free text, and as the development findings showed, free text says "out-of-context" for nearly every item. The pipeline was feeding the final verdict a hint that hardly ever changed, and the model followed it. The evidence itself made almost no difference (-0.011, $p = 0.619$).

Used as a separate check and combined with the plain verdict score, the evidence gave the best COSMOS result in the study: 0.923 AUROC and 84.0% accuracy. The gain over the plain score, however, is small and not significant ($+0.013$, 95% CI [-0.000, +0.028], $p = 0.052$). I left retrieval out of the final system.

9) Explanation quality: The explainer produced an explanation for all 256 test items, averaging 95 words. I rated a stratified sample of 50 from 1 to 5 on factual accuracy, coherence and relevance (Yao et al., 2023; Qi et al., 2024), without knowing which verdicts were correct. I opened the answer key only after scoring all 50 (Table VIII).

Factual accuracy scored clearly higher than coherence and relevance. The explanations usually describe the scene cor-

TABLE VIII
EXPLANATION RATINGS (1 TO 5)

	All (50)	Correct (43)	Wrong (7)	p
Factual accuracy	3.72	3.84	3.00	0.038
Coherence	3.04	3.19	2.14	0.014
Relevance	2.94	3.07	2.14	0.007

rectly, but they often skip the actual question of whether the two captions refer to the same event. Explanations of wrong verdicts were rated lower on every criterion (Mann-Whitney test, Mann and Whitney, 1947; $p = 0.007$ for the mean). Even so, an automatic check of all 256 found that 42 of the 43 wrong verdicts came with an explanation arguing for the wrong answer. A careful reader can often spot weak reasoning, but a confident explanation tells you very little about whether the verdict is right.

C. Support for the Research Question

The findings of this study clearly show that a domain-adapted 7B LVLM can detect out-of-context news images accurately: 83.2% accuracy and 0.910 AUROC. The gain from news-domain tuning is significant on COSMOS and on MMFakeBench, which the model never saw. The final system has no overlap with its test data, and it sits well above simple baselines.

The second half of the question, which components contribute, has a mixed answer. Reading the verdict as a calibrated probability and Stage 1 tuning both help. Combining separate questions helps on MMFakeBench. Stage 2 and the built-in retrieval pipeline do not help, and the explanations, though readable, are not reliable evidence that a verdict is right. The blank-image result tempers all of this, since the system gets most of its signal from the captions.

V. DISCUSSION

A. Interpretation of Results

The biggest change in performance came from reading the answer differently. Calibrated scoring gained 28.5 points on the untrained model, against 4.7 points from Stage 1. The probabilities were always ranking items sensibly. They were simply shifted so far upward that a 0.5 cut-off, which is effectively what free text uses, put nearly everything in one class.

Why did Stage 1 help if it never learned to recognise people? A likely reason is that it got the model used to how news captions are written. Almost half the named people in the data appear only once, and the tuned model could only name people who were already famous. I did not test this with a control model tuned on non-news captions, so it remains the most plausible account and not a proven one.

Stage 2's failure has a simpler cause. SNIFFER's task stage used 35,536 generated instructions. Mine had 1,189 labelled examples, and each target paired the verdict with one of just two fixed reasoning sentences. Stage 2 did push the model to use the image more, but accuracy fell. With so little varied data, it probably could not learn cross-modal reasoning, and its split shared images with the test set.

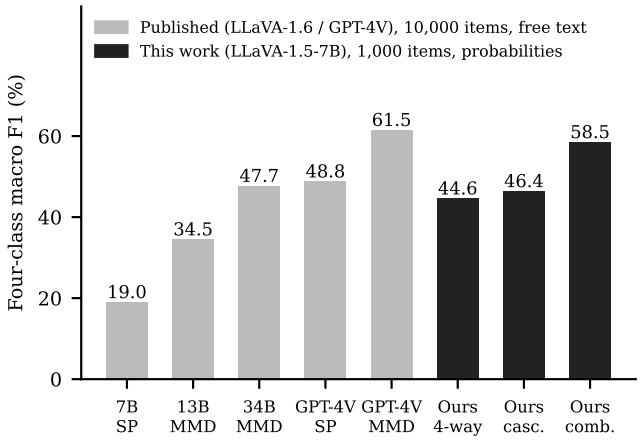


Fig. 3. Four-class macro F1 on MMFakeBench: published LLaVA-1.6 and GPT-4V results (SP: standard prompting; MMD: MMD-Agent) (Liu et al., 2025) and this work. Not a controlled comparison: this work uses a 1,000-item sample and the combination is fitted on 1,000 calibration items.

Three separate results point to text as the main signal: the blank-image ablation, the strong SBERT baseline and the MMFakeBench breakdown. Some of the headline accuracy therefore comes from patterns in COSMOS's captions. The visual check suggests the model is capable of more. It found edited images with an AUROC of 0.974 when asked directly, so the problem is the single verdict question, which never makes it look closely.

B. Comparison with Existing Literature

Table IX sets the results beside published work. Test sets and models differ in most cases, so these are reference points and not controlled comparisons.

On COSMOS, the final system comes within two points of the original COSMOS method (Aneja et al., 2023). It does this with a general-purpose LVLM and no object detector, though on a smaller test split. The blank-image result matches VERITE's finding (Papadopoulos et al., 2023) that COSMOS can largely be solved from text, now shown with an LVLM. My CLIP baseline on NewsCLIPpings (65.3%) is very close to the CLIP figure SNIFFER reports (66.0%), which gives me some confidence that the baseline is implemented correctly.

1) How this work differs from COSMOS: My COSMOS result is within two points of the original method, so it is worth being clear about how differently the two systems work (Table X). COSMOS is built from specialised parts: an object detector, a sentence encoder, a separately trained matching network and a hand-set rule with two thresholds. It locates the object each caption describes, so its decision is grounded in a specific region of the photo.

My system has none of those parts. There is no object detector, no separate caption encoder and no hand-written rule. One general-purpose model, LLaVA, reads the whole image

TABLE IX
COMPARISON WITH PUBLISHED WORK

Work	Approach	Reported result	This work
COSMOS (Aneja et al., 2023)	Self-supervised object-caption grounding, full 1,700-image test set	85% accuracy; SBERT baseline 77%	83.2% accuracy, 0.910 AUROC with a 7B LVLM on a 256-item labelled test split, without an object detector
VERITE (Papadopoulos et al., 2023)	Text-only vs multimodal models	Text-only best on COSMOS (72.6%)	Removing the image costs 0.014 AUROC (n.s.): confirms caption-driven detection with an LVLM
SNIFFER (Qi et al., 2024)	InstructBLIP/Vicuna-13B, two-stage tuning with 35,536 GPT-4 instructions; internal and external checks	88.4% on NewsCLIPpings; CLIP 66.0%; evidence adds 0.2 points	Stage 1 helps, Stage 2 with 1,189 examples does not; fused evidence adds 0.013 AUROC (n.s.); CLIP on NewsCLIPpings 65.3%
NewsCLIPpings (Luo et al., 2021)	CLIP ViT-B/32 classifier; human study	CLIP 60.2%; humans 65.6%	Zero-shot CLIP with validation threshold: 65.3%
LEMMA (Xuan et al., 2024)	GPT-4V with initial judgement, queries, retrieval, refinement	82.4% Twitter, 82.8% Fakeddit; removing initial stage lowers Twitter to 80.9%	Removing the initial assessment raises AUROC by 0.115: with a 7B model the free-text first guess is near-constant
Tahmasebi et al. (2024)	LLaVA-1.6-7B verification with retrieved or gold evidence	Retrieved F1 0.451 vs baseline 0.491; gold 0.534	Retrieved web evidence alone gives no measurable gain (-0.011, n.s.)
MMFakeBench (Liu et al., 2025)	LVLMs answering in free text; MMD-Agent decomposition	Four-class F1: LLaVA-1.6-7B 19.0; 34B + MMD-Agent 47.7; GPT-4V + MMD-Agent 61.5	LLaVA-1.5-7B read by probabilities: 44.6 (four-way prompt), 58.5 (learned combination, 1,000-item sample)

TABLE X
COSMOS AND THIS WORK COMPARED

	COSMOS (Aneja et al., 2023)	This work
Image model	Mask R-CNN, ResNet-50 backbone (frozen)	LLaVA-1.5's CLIP vision encoder (frozen)
Text model	Universal Sentence Encoder (frozen); SBERT at test time	LLaVA-1.5's Vicuna-7B, 4-bit, with LoRA
Training	Matching network, 160k unlabelled images	Stage 1 LoRA, 12k unlabelled image-caption pairs
Labels used	None	254 validation items (threshold)
Object grounding	Explicit, per detected object	None; whole image
Decision	Rule: IoU > 0.5 and SBERT similarity < 0.5	$P(\text{out-of-context}) > 0.81$
Test data	1,700 labelled images	256 of those items
Accuracy	85%	83.2% (AUROC 0.910)

and both captions together and gives a single probability. Its only threshold, 0.81, was chosen from validation data. I also used labels that COSMOS did not need: 254 validation items to choose the threshold. And the data differ. My 256 test items are a subset of the same 1,700 labelled items that COSMOS was tested on, because I used the rest for Stage 2 training and validation.

Why are the two results so close when the systems are so different? A likely reason is that both lean heavily on the captions. Half of COSMOS's rule is an SBERT check on whether the two captions say different things, and that check alone reaches 77%. My grey-image ablation shows that my system also gets most of its signal from the captions. So the two systems may land close together mainly because they pick up the same caption differences. That also means neither result, on its own, shows strong reasoning about whether the image and the captions match.

2) Other comparisons: The MMFakeBench comparison is the most striking (Fig. 3). The published result for 7B LLaVA under standard prompting is 19.0 macro F1. Reading the probabilities of a 7B LLaVA model, I got 44.6 from a single four-way question. The learned combination reached 58.5, above LLaVA-1.6-34B with MMD-Agent (47.7) and close to GPT-4V (61.5) (Liu et al., 2025). I would not call this a fair contest, though. My combination was fitted on 1,000 calibration items and evaluated on a 1,000-item sample, while the published numbers are zero-shot on the full test set.

Some of the design comparisons line up with the literature and some do not. SNIFFER found that evidence added 0.2 points (Qi et al., 2024). My fused evidence added 0.013 AUROC, which was not significant. Both suggest the internal check does most of the work on these benchmarks. LEMMA's first-judgement step helps GPT-4V (Xuan et al., 2024), since removing it lowered their Twitter accuracy from 82.4% to 80.9%. In my pipeline the same step did the opposite, because a 7B model's free-text first guess is nearly always the same. Tahmasebi et al. (2024) saw retrieved evidence fall below their baseline while gold evidence helped, which fits my finding that the retrieved evidence carried little signal.

C. Implications and Limitations of the Study

For anyone evaluating LVLM detectors, the main implication is that parsing free text can hide most of what the model can do, by a margin big enough to change the conclusion. Results based on probabilities, with thresholds set on validation data, AUROC and a calibration check, should be reported alongside free-text numbers. For anyone deploying such a detector, the raw probabilities should be recalibrated, and evidence works better as a separate check than inside the verdict prompt.

This study was limited by a small COSMOS test set of 256 items. My Stage 2 split was not grouped by image, so half the test set overlapped with Stage 2's training data, although the conclusions held on the clean subset. I compared MMFakeBench configurations on a 2,000-item sample, and I

could not run the LVLM on NewsCLIPpings, so the comparison with SNIFFER is indirect. I was the only person rating explanations, and I knew how the system worked. Finally, the explanation for Stage 1's gain was not tested with a control.

VI. Conclusion

A. Summary of Key Findings

With news-domain tuning and calibrated reading, an open 7B LVLM reached 83.2% accuracy and 0.910 AUROC on COSMOS. Its gain over the untrained model was significant on both COSMOS and MMFakeBench. Calibrated reading alone took the untrained model from 50.0% to 78.5%. The system relies mostly on the captions. Stage 2 and the integrated retrieval pipeline did not help, and mismatch detection on MMFakeBench stayed near chance. In 42 of 43 wrong verdicts, the explanation argued for the wrong answer.

B. Contributions to the Field

1) A low-cost detector (LLaVA-1.5-7B, 4-bit, LoRA, one free 16 GB GPU) that reaches 83.2% accuracy and 0.910 AUROC on held-out COSMOS data.
2) Evidence that free-text evaluation substantially understates LVLM detectors. Calibrated scoring raised the untrained model from 50.0% to 78.5% on COSMOS, and LLaVA's four-class macro F1 on MMFakeBench from a published 19.0 to 44.6.
3) A result checked for leakage, significance, generalisation to an unseen benchmark, reliance on the image, and simple baselines.
4) Evidence that LVLM detection on COSMOS is caption-driven, confirming VERITE's unimodal-bias finding with an LVLM.
5) An account of why LEMMA-style integrated retrieval fails with a small open model, and evidence that SNIFFER-style score fusion is a safer way to add evidence.

C. Recommendations for Future Research

The next step should be to split the labelled data by image so that task tuning and testing never share a picture. Stage 2 would likely benefit from a larger and more varied set of generated instructions, as in SNIFFER. Mismatch detection needs training signals that force the model to compare image and caption. In the retrieval pipeline, the first assessment should come from the calibrated score instead of free text. A full NewsCLIPpings evaluation would allow a direct comparison with SNIFFER. Several independent raters should score the explanations.

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APPENDIX

All code, configuration files and supporting documents for this project are available on GitHub at https://github.com/TebogoSeopa20/Honours-Research-Project. The repository contains:

- `Research_Proposal.pdf`: the approved research proposal.
- `src/misinfodet/`: the Python package, with data loaders for COSMOS, MMFakeBench and NewsCLIPpings; model loading and the LoRA training loop; the detection pipeline and all prompts; query generation, evidence retrieval and re-ranking; and evaluation metrics.
- `scripts/`: one script per experiment, from data preparation and training to each evaluation reported here.
- `configs/`: a YAML file for every configuration reported, including `diag_stage1_only.yaml` for the final system.
- `tests/`: unit tests for data preparation, prompts, label masking, metrics and retrieval parsing.
- `docs/`: dataset acquisition notes (`DATASETS.md`) and the experiment log (`EXPERIMENT_LOG.md`).
- `Proof_of_Research_Notebook.ipynb`: the Kaggle notebook used to run the experiments, with their outputs.

All experiments ran on Kaggle's free NVIDIA T4 GPUs. The datasets are not redistributed; `docs/DATASETS.md` explains how to obtain them.